

2 Sequences

Sequences arise naturally whenever we wish to approximate a quantity. For example, consider the decimal expansion of

$$\frac{1}{7} = 0.142857142857 \dots$$

If we successively retain one more decimal place, we obtain the sequence

$$0.1, 0.14, 0.142, 0.1428, 0.14285, \dots$$

Each term is obtained by keeping one additional decimal place of the decimal expansion of $\frac{1}{7}$, and hence provides a better approximation to $\frac{1}{7}$ than the previous one. For instance, if we require the approximation to be accurate up to 10^{-4} , then any term with at least four decimal places is sufficient.

The notion of a sequence and its convergence provides a rigorous mathematical framework for studying such approximations. Intuitively, a sequence (x_n) converges to a real number x if its terms become arbitrarily close to x as n becomes sufficiently large. In other words, given any prescribed level of accuracy, all terms of the sequence beyond a certain stage approximate x to that accuracy.

2.1 Sequences and their convergence

Definition 7. A sequence of real numbers is a function

$$f : \mathbb{N} \rightarrow \mathbb{R}.$$

For each $n \in \mathbb{N}$, we write

$$x_n = f(n),$$

and call x_n the n -th term of the sequence. The sequence is usually denoted by (x_n) or $\{x_n\}$, and may also be written as

$$(x_n)_n, \quad \{x_n\}_n, \quad \text{or} \quad (x_1, x_2, \dots).$$

Example 7. Some examples of sequences are given below.

1. Fix $c \in \mathbb{R}$ and define $x_n = c$ for all $n \in \mathbb{N}$. Then $(x_n) = (c, c, c, \dots)$. This is called a constant sequence.

2. Let $x_n = \frac{1}{n}$. Then $(x_n) = (1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n}, \dots)$.
3. Let $x_n = \frac{(-1)^{n+1}}{n}$. Then $(x_n) = (1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots, \frac{1}{2k-1}, -\frac{1}{2k}, \dots)$.
4. Let $x_n = (-1)^n$. Then $(x_n) = (-1, 1, -1, 1, \dots)$.
5. Let $x_n = n$. Then $(x_n) = (1, 2, 3, 4, \dots)$.
6. Let $x_n = 2^n$. Then $(x_n) = (2, 4, 8, 16, \dots)$.
7. Let $x_n = \frac{1}{2^n}$. Then $(x_n) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots)$.
8. Let $x_1 = x_2 = 1$ and define $x_n = x_{n-1} + x_{n-2}$ for all $n \geq 3$.
Then $(x_n) = (1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$.

Example 8. Let

$$x_1 = 0.1, \quad x_2 = 0.14, \quad x_3 = 0.142, \quad x_4 = 0.1428, \quad \dots,$$

where x_n is obtained by retaining the first n digits after the decimal point in the decimal expansion of

$$\frac{1}{7} = 0.142857142857 \dots$$

Observe that

$$\left| \frac{1}{7} - x_n \right| < \frac{1}{10^n}, \quad \forall n \in \mathbb{N}.$$

Thus, given any prescribed level of accuracy, we can choose n sufficiently large so that x_n approximates $\frac{1}{7}$ to that accuracy. This example motivates the notion of convergence, which we study next.

Definition 8. Let (x_n) be a sequence of real numbers and let $x \in \mathbb{R}$. We say that (x_n) converges to x if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

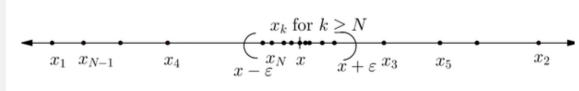
$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

In this case, x is called the limit of the sequence (x_n) . We write

$$x_n \rightarrow x \quad \text{as } n \rightarrow \infty,$$

or equivalently,

$$\lim_{n \rightarrow \infty} x_n = x.$$



In terms of quantifiers: (x_n) converges to x if

$$(\forall \varepsilon > 0) (\exists N \in \mathbb{N}) (\forall n \geq N) (|x - x_n| < \varepsilon).$$

Remark 1. 1. In the definition of convergence, the positive number ε may be chosen arbitrarily small. Nevertheless, for every $\varepsilon > 0$, there always exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

2. Let $\varepsilon > 0$. If (x_n) converges to x , then there exists $N \in \mathbb{N}$ such that

$$x_n \in (x - \varepsilon, x + \varepsilon), \quad \forall n \geq N.$$

Equivalently, the interval $(x - \varepsilon, x + \varepsilon)$ contains all but finitely many terms of the sequence (x_n) , i.e., only finitely many terms of the sequence can stay outside the interval.

Example 9. Consider the constant sequence $x_n = c$ for all $n \in \mathbb{N}$. We claim that

$$\lim_{n \rightarrow \infty} x_n = c.$$

Indeed, let $\varepsilon > 0$ be given. Since

$$|x_n - c| = 0 < \varepsilon, \quad \forall n \in \mathbb{N},$$

we may simply choose $N = 1$. Hence,

$$|x_n - c| < \varepsilon, \quad \forall n \geq N.$$

Therefore, (x_n) converges to c .

Example 10. Consider the sequence $(\frac{1}{n})$. We claim that

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Let $\varepsilon > 0$ be given. We have

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n}.$$

By the Archimedean Property, there exists $N \in \mathbb{N}$ such that

$$N > \frac{1}{\varepsilon}.$$

Hence, for every $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Exercise 8. Using the previous example, show that if $a, b \in \mathbb{R}$ with $a \neq 0$, then the sequence

$$x_n = \frac{1}{an + b}$$

converges. Determine its limit.

Example 11. Consider the sequence

$$a_n = \frac{n^2 + 1}{n^2}.$$

We claim that

$$\lim_{n \rightarrow \infty} a_n = 1.$$

Let $\varepsilon > 0$ be given. We have

$$|a_n - 1| = \left| \frac{n^2 + 1}{n^2} - 1 \right| = \frac{1}{n^2}.$$

By the Archimedean Property, there exists $N \in \mathbb{N}$ such that

$$N > \frac{1}{\sqrt{\varepsilon}}.$$

Hence, for every $n \geq N$,

$$|a_n - 1| = \frac{1}{n^2} \leq \frac{1}{N^2} < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{n^2 + 1}{n^2} = 1.$$

From the examples above, we see that proving the convergence of a sequence usually has two steps. First, we guess its limit. Then we show that the terms of the sequence get closer and closer to this value by estimating $|x_n - x|$. This raises two natural questions:

How do we guess the limit? & Can a sequence have more than one limit?

The first question becomes easier with practice. We now answer the second question.

Theorem 5 (Uniqueness of the Limit). *If*

$$\lim_{n \rightarrow \infty} x_n = x \quad \text{and} \quad \lim_{n \rightarrow \infty} x_n = y,$$

then $x = y$.

Proof. Let $\varepsilon > 0$ be given. Since $x_n \rightarrow x$, there exists $N_1 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{2}, \quad \forall n \geq N_1.$$

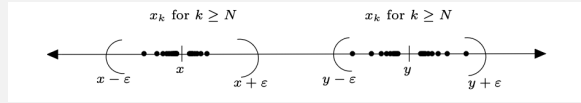
Similarly, since $x_n \rightarrow y$, there exists $N_2 \in \mathbb{N}$ such that

$$|x_n - y| < \frac{\varepsilon}{2}, \quad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$, the triangle inequality gives

$$|x - y| = |x - x_n + x_n - y| \leq |x - x_n| + |x_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $x = y$.



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